Elementary Problems
Counting
Solutions

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Basic Work

1. Consider an arbitrary person A_1. There are 9 ways to find a partner, say A_2, for A_1. After removing A_1 and A_2, we consider an arbitrary person A_3. There are 7 ways to find a partner, say A_4, for A_3. Similarly, there are 5 ways to find a partner for a remaining person A_5, and 3 ways to find a partner for another person A_7. After that, the two remaining people must form a pair. So the answer is $9 \times 7 \times 5 \times 3 = 945$.

2. To form a subset of S, each of the n elements can be chosen or not, and such a choice is independent from any other choices. Hence, there are 2^n subsets of S, where $2^n - 1$ of them are nonempty.

3. Since the sets are nonempty, we need $n \geqslant 3$. WLOG assume $1 \in S_1$ and $2 \in S_2$. Starting from 3, each integer k cannot belong to the same set as $k-1$. There are 2 ways to put k into a set. So there are 2^{n-2} ways to place all the numbers to avoid two consecutive integers belonging to the same set. However, we need to subtract those cases for which some sets are empty. Indeed, as $1 \in S_1$ and $2 \in S_2$, the only possibility is that the remaining set is empty. In that case, there is a unique way to put the integers $3, 4, \ldots, n$. So the answer is $2^{n-2} - 1$.

4. When $n = 1$, there is clearly 1 way to do so. When $n \geqslant 2$, we first fill in the top left $(n-1) \times (n-1)$ board arbitrarily in $n^{(n-1)^2}$ ways. We claim that there is a unique way to fill in the remaining cells so that the condition is satisfied.

   We use a_{ij} to denote the cell in the ith row and the jth column. For the 1st row, we need
   $$a_{11} + a_{12} + \cdots + a_{1(n-1)} + a_{1n} \equiv 0 \pmod{n}.$$
   Since the first $n-1$ numbers are fixed, the residue class of a_{1n} is fixed. As a_{1n} can be one of $1, 2, \ldots, n$, there is exactly one choice for which the sum of numbers in the first row is divisible by n. Similarly, by considering the sum of numbers in the 2nd row to the $(n-1)$st row, we can determine a_{in} uniquely where $2 \leqslant i \leqslant n-1$. Also, by considering the sum of numbers in the 1st column to the nth column, we can determine a_{nj} uniquely where $1 \leqslant j \leqslant n$. All conditions are satisfied, possibly except the sum of the nth row.

   Let r_i be the sum of numbers in the ith row and let c_j be the sum of numbers in the jth column. Note that
   $$r_1 + r_2 + \cdots + r_n = c_1 + c_2 + \cdots + c_n$$
   since both sides represent the sum of all numbers in the table. As we know that n divides each of $r_1, r_2, \ldots, r_{n-1}$ and $c_1, c_2, \ldots, c_n$, we obtain $n \mid r_n$. So all the conditions are satisfied.

   Thus, the number of ways to fill in the table is $n^{(n-1)^2}$, which also includes the case $n = 1$.

5. **Solution 1:**

   Let the subset be $\{a, b, c\}$ where $a < b < c$. For even $n = 2k$, the cases $b \leqslant k$ and $b \geqslant k+1$ are symmetric. Suppose $b \leqslant k$. Then each choice of a for which $a < b$ gives a unique choice of c such that $b - a = c - b$. Note that we must have

   $$c = 2b - a \leqslant 2k - 1 < n.$$

   So it is the same as choosing 2 elements a, b from 1 to k. This gives $\binom{k}{2} = \dfrac{k(k-1)}{2}$ subsets for $b \leqslant k$. Hence, the answer is $k(k-1) = \dfrac{n(n-2)}{4}$ subsets when n is even.

   For odd $n = 2k+1$, the cases $b \leqslant k$ and $b \geqslant k+2$ are symmetric. As above, there are $2\binom{k}{2} = k(k-1)$ subsets in these cases. For $b = k+1$, each of the k choices of a gives a unique choice of c. Hence, the answer is

   $$k(k-1) + k = k^2 = \frac{(n-1)^2}{4}$$

   subsets when n is odd.

   Solution 2:

   In order that $b - a = c - b$, i.e. $a + c = 2b$, a and c must have the same parity. Conversely, for any pair of (a, c) with the same parity, we can always find a unique b such that $b = \dfrac{a+c}{2}$, which clearly lies between 1 and n if both a and c do. So it is the same as finding the number of pairs of (a, c) with the same parity.

   For even n, there are $\dfrac{n}{2}$ odd numbers and $\dfrac{n}{2}$ even numbers. So the answer is

   $$2\binom{\frac{n}{2}}{2} = \frac{n(n-2)}{4}$$

   subsets when n is even.

   For odd n, there are $\dfrac{n+1}{2}$ odd numbers and $\dfrac{n-1}{2}$ even numbers. So the answer is

   $$\binom{\frac{n+1}{2}}{2} + \binom{\frac{n-1}{2}}{2} = \frac{(n+1)(n-1)}{8} + \frac{(n-1)(n-3)}{8} = \frac{(n-1)^2}{4}$$

   subsets when n is odd.

6. There are $\left\lfloor \dfrac{n}{3} \right\rfloor$ elements of the form $3k$, $\left\lfloor \dfrac{n+1}{3} \right\rfloor$ elements of the form $3k+1$, and $\left\lfloor \dfrac{n+2}{3} \right\rfloor$ elements of the form $3k+2$.

The remainders of the 3 elements in the subset may be pairwise the same or pairwise distinct. For the former case, there are

$$\binom{\lfloor \frac{n}{3} \rfloor}{3} + \binom{\lfloor \frac{n+1}{3} \rfloor}{3} + \binom{\lfloor \frac{n+2}{3} \rfloor}{3}$$

subsets. For the latter case, there are

$$\left\lfloor \frac{n}{3} \right\rfloor \left\lfloor \frac{n+1}{3} \right\rfloor \left\lfloor \frac{n+2}{3} \right\rfloor$$

subsets.

For $n = 3k$, the answer is

$$3\binom{k}{3} + k^3 = \frac{k(3k^2 - 3k + 2)}{2}.$$

For $n = 3k + 1$, the answer is

$$2\binom{k}{3} + \binom{k+1}{3} + k^2(k+1) = \frac{k(3k^2 + 1)}{2}.$$

For $n = 3k + 2$, the answer is

$$\binom{k}{3} + 2\binom{k+1}{3} + k(k+1)^2 = \frac{k(3k^2 + 3k + 2)}{2}.$$

7. There are $\binom{n^2}{2} = \frac{n^2(n^2 - 1)}{2}$ ways to put 2 kings on the chessboard. We count the number of ways for which the kings threaten each other.

   Firstly, if the kings lie on two adjacent cells in the same row, there are n ways to choose a row and $n - 1$ ways to choose the position for the king on the left. There are $n(n - 1)$ possibilities. By symmetry, there are $n(n - 1)$ possibilities if they lie on two adjacent cells in the same column.

   Secondly, suppose the kings lie on two adjacent cells in the same diagonal parallel to the main diagonal. There are $1, 2, 3, \ldots, n, n - 1, \ldots, 1$ cells in the diagonals parallel to the main diagonal respectively. For such a diagonal with k cells, there are $k - 1$ ways to put the kings in adjacent positions. So there are

$$0 + 1 + 2 + \cdots + (n - 1) + (n - 2) + \cdots + 0 = (n - 1)^2$$

   possibilities. By symmetry, there are $(n - 1)^2$ possibilities if they lie on two adjacent cells in the same diagonal of the other direction. Thus, the answer is

$$\frac{n^2(n^2 - 1)}{2} - 2n(n - 1) - 2(n - 1)^2 = \frac{(n - 1)(n - 2)(n^2 + 3n - 2)}{2}.$$

8. Suppose the digits are A, A, B, B, C. Then the number can only be $AABBC$, $AACBB$ or $CAABB$. For each case, the first digit cannot be 0. There are 9 choices. Afterwards, there are 9 and 8 choices respectively to choose the remaining two digits. There are

$$3 \times 9 \times 9 \times 8 = 1944$$

such numbers.

9. The number of integers divisible by 3 is $\left\lfloor \frac{N}{3} \right\rfloor$. By the inclusion-exclusion principle, the number of integers divisible by 5 or 7 is $\left\lfloor \frac{N}{5} \right\rfloor + \left\lfloor \frac{N}{7} \right\rfloor - \left\lfloor \frac{N}{35} \right\rfloor$. Hence, we need

$$\left\lfloor \frac{N}{3} \right\rfloor = \left\lfloor \frac{N}{5} \right\rfloor + \left\lfloor \frac{N}{7} \right\rfloor - \left\lfloor \frac{N}{35} \right\rfloor.$$

Let $N = 35k + r$ where $0 \leqslant r \leqslant 34$. Note that $\left\lfloor \frac{N}{3} \right\rfloor \geqslant \frac{N-2}{3}$. Hence we get

$$\frac{(35k+r)-2}{3} \leqslant 7k + \frac{r}{5} + 5k + \frac{r}{7} - k = 11k + \frac{12r}{35}.$$

This implies $35(2k + r - 2) \leqslant 36r$, i.e. $70k \leqslant r + 70$. As $r < 35$, we must have $k \leqslant 1$. Thus, $N \leqslant 69$.

- When $N = 69$, the left-hand side is 23 while the right-hand side is $13 + 9 - 1 = 21$.
- When $N = 68, 67, 66$, the left-hand side is 22 while the right-hand side is 21.
- When $N = 65$, the left-hand side is 21 while the right-hand side is 21.

So the largest positive integer N is 65.

10. For each positive integer k, let A_k be the set of integers from 1 to 1000 which are divisible by k. Then $|A_k| = \left\lfloor \frac{1000}{k} \right\rfloor$. By the inclusion-exclusion principle, we have

$$\begin{aligned}
&|A_4 \cup A_5 \cup A_6| \\
&= |A_4| + |A_5| + |A_6| - |A_4 \cap A_5| - |A_5 \cap A_6| - |A_6 \cap A_4| + |A_4 \cap A_5 \cap A_6| \\
&= |A_4| + |A_5| + |A_6| - |A_{20}| - |A_{30}| - |A_{12}| + |A_{60}| \\
&= \left\lfloor \frac{1000}{4} \right\rfloor + \left\lfloor \frac{1000}{5} \right\rfloor + \left\lfloor \frac{1000}{6} \right\rfloor - \left\lfloor \frac{1000}{20} \right\rfloor - \left\lfloor \frac{1000}{30} \right\rfloor - \left\lfloor \frac{1000}{12} \right\rfloor + \left\lfloor \frac{1000}{60} \right\rfloor \\
&= 250 + 200 + 166 - 50 - 33 - 83 + 16 \\
&= 466.
\end{aligned}$$

The positive integers not divisible by $4, 5$ or 6 are those which do not belong to the union $A_4 \cup A_5 \cup A_6$. So the answer is $1000 - 466 = 534$.

11. Let A_i be the set of rearrangements for which person i takes the original seat. Then $|A_i|$ counts those permutations of the remaining 4 people, and hence $|A_i| = 4! = 24$. Similarly, we have $|A_i \cap A_j| = 3! = 6$ for $i \neq j$, etc. By the inclusion-exclusion principle, we get

$$\begin{aligned}|A_1 \cup A_2 \cup \cdots \cup A_5| &= \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| \\ &\quad - \sum_{i<j<k<\ell} |A_i \cap A_j \cap A_k \cap A_\ell| + |A_1 \cap A_2 \cap \cdots \cap A_5| \\ &= 5 \times 24 - \binom{5}{2} \times 6 + \binom{5}{3} \times 2 - \binom{5}{4} \times 1 + 1 \\ &= 120 - 60 + 20 - 5 + 1 = 76.\end{aligned}$$

As there are $5! = 120$ rearrangements in total, there are $120 - 76 = 44$ rearrangements which do not belong to any of the A_i's. This means none of the people takes the original seat.

12. Let the regular polygon be $A_1A_2\cdots A_n$. We count the number of isosceles triangles with A_1 as the vertex opposite to the base. The isosceles triangles are $\triangle A_1A_2A_n$, $\triangle A_1A_3A_{n-1}$, …, $\triangle A_1A_sA_t$, where $s = \dfrac{n+1}{2}$ and $t = \dfrac{n+3}{2}$ if n is odd, and $s = \dfrac{n}{2}$ and $t = \dfrac{n+4}{2}$ if n is even. By symmetry, the number of isosceles triangles with any other A_j as the vertex opposite to the base is the same.

If n is not a multiple of 3, then there is no equilateral triangle formed by the vertices. So each isosceles triangle is counted exactly once. There are $\dfrac{n(n-1)}{2}$ isosceles triangles if n is odd, and $\dfrac{n(n-2)}{2}$ isosceles triangles if n is even.

If n is a multiple of 3, then there are $\dfrac{n}{3}$ equilateral triangles. Each such equilateral triangle is counted 3 times in the above step. Therefore, there are

$$\frac{n(n-1)}{2} - 2 \cdot \frac{n}{3} = \frac{n(3n-7)}{6}$$

isosceles triangles if n is of the form $6k+3$, and

$$\frac{n(n-2)}{2} - 2 \cdot \frac{n}{3} = \frac{n(3n-10)}{6}$$

isosceles triangles if n is a multiple of 6.

13. We first count the number of nonempty subsets A such that $3 \mid S(A)$. If we choose k elements in $\{1, 4, 7\}$ where $k = 1$ or 2, then we need to choose k elements in $2, 5, 8$. If we choose 0 or 3 elements in $\{1, 4, 7\}$, then we can choose 0 or 3 elements in $\{2, 5, 8\}$.

Afterwards, for each of 3 and 6, we can either choose it or not. This forms a subset with sum of elements being a multiple of 3, but it may not be nonempty. So there are

$$(3 \times 3 + 3 \times 3 + 2 \times 2) \times 2 \times 2 - 1 = 87$$

nonempty subsets A with $3 \mid S(A)$. It remains to remove those A for which $15 \mid S(A)$. We can simply list out all these subsets as follows.

$$\begin{aligned}
\{7,8\},& \\
\{4,5,6\}, \{3,5,7\}, \{3,4,8\}, \{2,5,8\}, \{2,6,7\}, \{1,6,8\},& \\
\{2,3,4,6\}, \{1,3,5,6\}, \{1,3,4,7\}, \{1,2,5,7\}, \{1,2,4,8\},& \\
\{1,2,3,4,5\},& \\
\{4,5,6,7,8\},& \\
\{2,3,4,6,7,8\}, \{1,3,5,6,7,8\},& \\
\{1,2,3,4,5,7,8\}.&
\end{aligned}$$

There are 17 such subsets. So the answer is $87 - 17 = 70$.

14. We use W and B to denote white and black cells respectively. We first consider those colourings of a $k \times 8$ board for which row 1 is $WBWBWBWB$. We claim that the colouring is uniquely determined by the first cells of the remaining rows.

    Indeed, if the first cell in row 2 is W, then row 2 must be $WBWBWBWB$ by using the condition. If the first cell in row 2 is B, then row 2 must be $BWBWBWBW$. We can prove inductively that the colouring of the remaining rows are determined uniquely by their first cells.

    For the original problem, if row 1 is $WBWBWBWB$ or $BWBWBWBW$, the above arguments show that there are $2^7 = 128$ such colourings. If row 1 is not of the above types, then there must be two consecutive cells having the same colour, say WW. Note that the two cells beneath them are BB, and inductively, we can work out all cells in these two columns. The following board shows an example.

		W	W				
		B	B				
		W	W				
		B	B				
		W	W				
		B	B				
		W	W				
		B	B				

These two columns partition the board into two parts (probably with an empty part), each of which is of size $8 \times k$ for some k. From our observation at the beginning, the colouring of the remaining table is uniquely determined by row 1. So there is a unique colouring for each of these $2^8 - 2 = 254$ cases.

Therefore, the number of colourings is $128 + 128 + 254 = 510$.

<u>Permutation and Combination</u>

15. Since the first digit is nonzero, there are 3 ways to choose the first digit. Suppose the first digit is 1. The other 1 can be put in one of the remaining 7 places. Next, there are $\binom{6}{2}$ ways to choose two places for the 0's, and then $\binom{4}{2}$ ways to choose two places for the 2's. This determines the 8-digit number. So the answer is

$$3 \times 7 \times \binom{6}{2} \times \binom{4}{2} = 1890.$$

16. Let the numbers be $(a_1, a_2, \ldots, a_6)$ in the anticlockwise direction. We may assume $a_1 = 0$. From the convention, we may assume the other 0 is a_2, a_3 or a_4. If $a_2 = 0$ or $a_3 = 0$, then 2 of the remaining 4 numbers are 1 and the configuration is uniquely determined. There are $\binom{4}{2} = 6$ choices for each case.

    If $a_4 = 0$, then we have $(0, 1, 1, 0, 2, 2)$, $(0, 1, 2, 0, 1, 2)$, $(0, 1, 2, 0, 2, 1)$ or $(0, 2, 1, 0, 2, 1)$. Hence, the total number of ways is

$$2 \times 6 + 4 = 16.$$

17. **Solution 1:**

    Suppose 9 is the kth term, where $k = 2, 3, \ldots, 8$. It suffices to choose $k - 1$ terms from 1 to 8 and put them in increasing order before 9, and put the remaining numbers in decreasing order after 9. So the answer is

$$\binom{8}{1} + \binom{8}{2} + \cdots + \binom{8}{7} = 8 + 28 + 56 + 70 + 56 + 28 + 8 = 254.$$

    Solution 2:

    For each of $1, 2, \ldots, 8$, we can put it before or after 9. Afterwards, we can order the numbers before 9 and the numbers after 9 in a unique way so that the sequence is first increasing and then decreasing. There are $2^8 = 256$ ways to rearrange the numbers. However, 2 cases should be rejected since they are entirely increasing or decreasing. So the answer is $256 - 2 = 254$.

18. There are $8 \times 8 = 64$ ways to choose a pair of a row and a column. This pair consists of 15 cells and we need to put 10 chess pieces on some of these cells. Since $10 > 8$, these chess pieces cannot lie on the same row or the same column. So each admissible case is counted exactly once. Hence, there are

$$64 \times \binom{15}{10} = 192192$$

ways to put the chess pieces.

19. If one of the digits is 0, we may assume the digits are 0 and 1. For a k-digit number, the leading digit must be 1, and there are 2 choices for each of the remaining $k-1$ digits. As k ranges from 1 to 5, there are

$$1 + 2 + 2^2 + 2^3 + 2^4 = 31$$

integers not exceeding 99999 which consist of the digits 0 and 1 only. However, there are 5 of them $1, 11, \ldots, 11111$ which do not consist of 0. So there are

$$\binom{31}{3} - \binom{5}{3} = 4485$$

ways to choose 3 integers which only consist of the digits 0 and 1.

If both digits are nonzero, we may assume the digits are 1 and 2. For a k-digit number, there are 2 choices for each digit. So there are

$$2 + 2^2 + 2^3 + 2^4 + 2^5 = 62$$

integers not exceeding 99999 which consist of the digits 1 and 2 only. As above, we need to reject those cases which consist of only 1 digit. There are

$$\binom{62}{3} - \binom{5}{3} - \binom{5}{3} = 37800$$

ways to choose 3 integers which only consist of the digits 1 and 2.

Hence, the answer is

$$4485 \times 9 + 37800 \times \binom{9}{2} = 1401165.$$

20. There are $\binom{n}{k}$ ways to choose k rows to put the rooks, and there are P_k^n ways to put these rooks in those rows so that no two occupy the same column since we have to choose k columns in total and the order matters. Thus, the answer is

$$\binom{n}{k} P_k^n = \frac{(n!)^2}{k!((n-k)!)^2}.$$

21. (a) There are $\binom{n}{3}$ ways to choose the subset $\{f(1), f(2), f(3)\}$. Each choice gives a unique way to assign the images of $1, 2, 3$. Also, there are n choices for each of $f(k)$ where $k \geqslant 4$. So there are

$$\binom{n}{3} \cdot n^{n-3} = \frac{n^{n-2}(n-1)(n-2)}{6}$$

such functions.

(b) A bijective function from S to S is a permutation of the numbers $1, 2, \ldots, n$. We can first choose 3 numbers from the n numbers and place them in the first three positions in ascending order (which correspond to $g(1), g(2), g(3)$). Then the remaining $n-3$ numbers can be permuted arbitrarily. There are

$$\binom{n}{3} \cdot (n-3)! = \frac{n!}{6}$$

such functions.

22. We first regard the subsets to be distinct. Since each element may belong to one of the k subsets, there are k^n ways to partition the set $S = \{1, 2, \ldots, n\}$ into k subsets (not necessarily nonempty).

Let S_j be the number of partitions of S into k subsets such that the jth subset is empty. Note that

$$|S_{i_1} \cap S_{i_2} \cap \cdots \cap S_{i_m}| = (k-m)^n$$

for $1 \leqslant i_1 < i_2 < \cdots < i_m \leqslant k$ since each element may belong to one of the remaining $k-m$ subsets. By the inclusion-exclusion principle, the number of partitions of S into k nonempty subsets is

$$\begin{aligned} & k^n - \sum_{i_1=1}^{k} |S_{i_1}| + \sum_{1 \leqslant i_1 < i_2 \leqslant k} |S_{i_1} \cap S_{i_2}| - \cdots + (-1)^k |S_1 \cap S_2 \cap \cdots \cap S_k| \\ = {} & k^n - \binom{k}{1}(k-1)^n + \binom{k}{2}(k-2)^n - \cdots + (-1)^{k-1}\binom{k}{k-1}1^n. \end{aligned}$$

As the subsets are indistinguishable, the answer is

$$\frac{1}{k!}\left(k^n - \binom{k}{1}(k-1)^n + \binom{k}{2}(k-2)^n - \cdots + (-1)^{k-1}\binom{k}{k-1}1^n\right).$$

23. Let a_k be the number of ways to climb a ladder of k steps. Suppose $k \geqslant 2$. If we climb 1 step in the last time, then we first need to climb $k-1$ steps before that. There are a_{k-1} ways to do so. If we climb 2 steps in the last time, then we first need to climb $k-2$ steps before that. There are a_{k-2} ways to do so. This yields

$$a_k = a_{k-1} + a_{k-2}.$$

Its characteristic equation is $\lambda^2 - \lambda - 1 = 0$. Its roots are $\dfrac{1 \pm \sqrt{5}}{2}$. Thus, we have $a_n = A\Big(\dfrac{1+\sqrt{5}}{2}\Big)^n + B\Big(\dfrac{1-\sqrt{5}}{2}\Big)^n$ for some constants A, B. One easily finds that $a_0 = 1$ and $a_1 = 1$. Solving the system

$$\begin{cases} A + B = 1, \\ A\Big(\dfrac{1+\sqrt{5}}{2}\Big) + B\Big(\dfrac{1-\sqrt{5}}{2}\Big) = 1, \end{cases}$$

we obtain $A = \dfrac{1+\sqrt{5}}{2\sqrt{5}}$ and $B = \dfrac{-1+\sqrt{5}}{2\sqrt{5}}$. Hence,

$$a_n = \frac{1}{\sqrt{5}}\Big(\Big(\frac{1+\sqrt{5}}{2}\Big)^{n+1} - \Big(\frac{1-\sqrt{5}}{2}\Big)^{n+1}\Big).$$

24. Let a_n be the number of such bijections. Clearly, $a_1 = 1$ and $a_2 = 2$. Consider $n \geqslant 3$. If $f(n) = n$, then the restriction of f on the set $\{1, 2, \ldots, n-1\}$ is a bijection with the same condition being satisfied. So there are a_{n-1} such bijections. If $f(n) = n-1$, then $f(n-1) = n$ since otherwise there is no number mapping to n. In that case, the restriction of f on the set $\{1, 2, \ldots, n-2\}$ is a bijection with the same condition being satisfied. So there are a_{n-2} such bijections. Hence, we find that

$$a_n = a_{n-1} + a_{n-2}$$

if $n \geqslant 3$. Its characteristic equation is $\lambda^2 - \lambda - 1 = 0$. Its roots are $\dfrac{1 \pm \sqrt{5}}{2}$. Thus, we have $a_n = A\Big(\dfrac{1+\sqrt{5}}{2}\Big)^n + B\Big(\dfrac{1-\sqrt{5}}{2}\Big)^n$ for some constants A, B. As $a_1 = 1$ and $a_2 = 2$, we may further define $a_0 = 1$ so that $a_2 = a_1 + a_0$. Solving the system

$$\begin{cases} A + B = 1, \\ A\Big(\dfrac{1+\sqrt{5}}{2}\Big) + B\Big(\dfrac{1-\sqrt{5}}{2}\Big) = 1, \end{cases}$$

we obtain $A = \dfrac{1+\sqrt{5}}{2\sqrt{5}}$ and $B = \dfrac{-1+\sqrt{5}}{2\sqrt{5}}$. Hence,

$$a_n = \frac{1}{\sqrt{5}}\Big(\Big(\frac{1+\sqrt{5}}{2}\Big)^{n+1} - \Big(\frac{1-\sqrt{5}}{2}\Big)^{n+1}\Big).$$

25. **Solution 1:**

Let a_k and b_k be the number of routes of k steps which end at A and B respectively. Note that b_k also counts the number of routes of k steps which end at C (or D) due to symmetry.

For a route of k steps ending at A where $k \geqslant 1$, the previous step must end at B, C or D. So we have to go to one of B, C, D in $k-1$ steps and then return to A. There are $3b_{k-1}$ such routes by definition. Hence,

$$a_k = 3b_{k-1}. \tag{1}$$

Similarly, for a route of k steps ending at B where $k \geqslant 1$, the previous step must end at A, C or D. If the previous step ends at A, there are a_{k-1} such routes. If the previous steps ends at one of C, D, there are $2b_{k-1}$ such routes. Hence,

$$b_k = a_{k-1} + 2b_{k-1}. \tag{2}$$

Substituting (1) into (2), we get

$$\frac{1}{3}a_{k+1} = a_{k-1} + \frac{2}{3}a_k.$$

This means $a_{k+1} = 2a_k + 3a_{k-1}$. Its characteristic equation is $\lambda^2 - 2\lambda - 3 = 0$. Its roots are $3, -1$. Thus, we have $a_n = A \cdot 3^n + B(-1)^n$ for some constants A, B. One easily finds that $a_0 = 1$ and $a_1 = 0$. Solving the system

$$\begin{cases} A + B = 1, \\ 3A - B = 0, \end{cases}$$

we obtain $A = \dfrac{1}{4}$ and $B = \dfrac{3}{4}$. Hence,

$$a_n = \frac{3^n + 3(-1)^n}{4}.$$

Solution 2:

The total number of routes of k steps is 3^k since there are 3 choices for each step. Let a_k be the number of routes of k steps which end at A. Consider the position after $k-2$ steps.

If we end at A after $k-2$ steps, there are 3 choices for the $(k-1)$st step, and we can always return to A in the last step. If we end at B (or C or D) after $k-2$ steps, there are 2 choices, either C or D, for the $(k-1)$st step for which we can return to A in the last step. Note that there are $3^{k-2}-a_{k-2}$ ways for us to go to one of B, C, D after $k-2$ steps. This sets up a recurrence relation

$$a_k = 3a_{k-2} + 2(3^{k-2} - a_{k-2}) = 2 \cdot 3^{k-2} + a_{k-2}.$$

It follows easily that

$$\begin{aligned} a_n &= 2 \cdot 3^{n-2} + a_{n-2} = 2 \cdot 3^{n-2} + 2 \cdot 3^{n-4} + a_{n-4} \\ &= \cdots = 2(3^{n-2} + 3^{n-4} + \cdots + 3^r) + a_r \end{aligned}$$

where $r = 0$ if n is even and $r = 1$ if n is odd. Clearly, we have $a_0 = 1$ and $a_1 = 0$.

If n is even, then

$$a_n = 2 \cdot \frac{3^n - 1}{3^2 - 1} + a_0 = \frac{3^n - 1}{4} + 1 = \frac{3^n + 3}{4}.$$

If n is odd, then

$$a_n = 2 \cdot \frac{3^n - 3}{3^2 - 1} + a_1 = \frac{3^n - 3}{4}.$$

26. Let a_n be the number of such integers. By symmetry, there are exactly $\dfrac{a_n}{3}$ such numbers starting with a particular digit.

For $n \geqslant 3$, we may assume the first digit is 1. If the second digit is 2 or 3, then there are $\dfrac{2a_{n-1}}{3}$ possibilities for the second to the last digits, and each case gives exactly one n-digit number by putting the digit 1 at the beginning. If the second digit is 1, then the third digit is 2 or 3. Again, there are $\dfrac{2a_{n-2}}{3}$ possibilities for the second to the last digits. As the first digit can have 3 possibilities, we find that

$$a_n = 3\Big(\frac{2a_{n-1}}{3} + \frac{2a_{n-2}}{3}\Big) = 2a_{n-1} + 2a_{n-2}.$$

Its characteristic equation is $\lambda^2 - 2\lambda - 2 = 0$. Its roots are $1 \pm \sqrt{3}$. Thus, we have $a_n = A(1+\sqrt{3})^n + B(1-\sqrt{3})^n$ for some constants A, B. One easily finds that $a_1 = 3$ and $a_2 = 9$. We may further define $a_0 = \dfrac{3}{2}$ so that $a_2 = 2a_1 + 2a_0$. Solving the system

$$\begin{cases} A + B = \frac{3}{2}, \\ A(1+\sqrt{3}) + B(1-\sqrt{3}) = 3, \end{cases}$$

we obtain $A = \dfrac{3+\sqrt{3}}{4}$ and $B = \dfrac{3-\sqrt{3}}{4}$. Hence,

$$a_n = \frac{3+\sqrt{3}}{4}(1+\sqrt{3})^n + \frac{3-\sqrt{3}}{4}(1-\sqrt{3})^n.$$

27. Let a_n be the number of such integers whose first two digits are the same. Let b_n be the number of such integers whose first two digits are distinct. By convention, we should have $a_1 = 3$ and $b_1 = 0$. We need to determine $s_n = a_n + b_n$.

    Firstly, for a_n, by removing the first digit, the remaining $n-1$ digits also satisfy the condition. There are $a_{n-1} + b_{n-1}$ possibilities. Each possibility uniquely determines the n-digit number since the first two digits are the same, and the resultant first three digits cannot be pairwise distinct. So we have

$$a_n = a_{n-1} + b_{n-1} = s_{n-1}.$$

    Secondly, for b_n, we again ignore the first digit. If the second and the third digits are the same, there are 2 choices for the first digit so that it is different from the second. If the second and the third digits are different, there is only one choice for the first digit so that the first three digits are not pairwise distinct. This yields

$$b_n = 2a_{n-1} + b_{n-1} = s_{n-1} + a_{n-1} = s_{n-1} + s_{n-2}.$$

    We find that

$$s_n = a_n + b_n = 2s_{n-1} + s_{n-2}.$$

    Its characteristic equation is $\lambda^2 - 2\lambda - 1 = 0$. Its roots are $1 \pm \sqrt{2}$. Thus, we have $s_n = A(1+\sqrt{2})^n + B(1-\sqrt{2})^n$ for some constants A, B. One easily finds that $s_1 = 3$ and $s_2 = 9$. We may further define $s_0 = 3$ so that $s_2 = 2s_1 + s_0$. Solving the system

$$\begin{cases} A + B = 3, \\ A(1+\sqrt{2}) + B(1-\sqrt{2}) = 3, \end{cases}$$

    we obtain $A = B = \dfrac{3}{2}$. Hence,

$$s_n = \frac{3}{2}((1+\sqrt{2})^n + (1-\sqrt{2})^n).$$

28. Let a_n be the number of words with n digits that begin with 0. Then a_n is also the number of words with n digits that begin with 4. Let b_n be the number of words with n digits that begin with 1. Then b_n is also the number of words with n digits that begin with 3. Let c_n be the number of words with n digits that begin with 2.

    If a word with n digits begins with 0, then the remaining $n-1$ digits must form a word which begins with 1. Thus,

$$a_{n+1} = b_n.$$

    If a word with n digits begins with 1, then the remaining $n-1$ digits must form a word which begins with 0 or 2. Thus,

$$b_{n+1} = a_n + c_n.$$

If a word with n digits begins with 2, then the remaining $n-1$ digits must form a word which begins with 1 or 3. Thus,

$$c_{n+1} = 2b_n.$$

From the recurrence relations, for $n \geqslant 2$, we have

$$b_{n+1} = a_n + c_n = b_{n-1} + 2b_{n-1} = 3b_{n-1}.$$

This implies

$$a_{n+1} = b_n = 3b_{n-2} = 3a_{n-1}$$

and

$$c_{n+1} = 2b_n = 6b_{n-2} = 3c_{n-1}$$

for $n \geqslant 3$. This shows the total number s_n of words with n digits satisfies

$$s_{n+2} = 2a_{n+2} + 2b_{n+2} + c_{n+2} = 3(2a_n + 2b_n + c_n) = 3s_n$$

for $n \geqslant 2$.

One easily finds that $s_1 = 5$, $s_2 = 8$ and $s_3 = 14$. It follows from $s_{n+2} = 3s_n$ that

$$s_{2k} = 8 \cdot 3^{k-1} \quad \text{and} \quad s_{2k+1} = 14 \cdot 3^{k-1}$$

for $k \geqslant 1$.

29. Let the octagon be $ABCDEFGH$. Note that the odd steps can only go to B, D, F, H while the even steps can only go to A, C, E, G. Thus, we have $a_{2n-1} = 0$. Let b_n be the number of paths of exactly $2n$ jumps ending at A without reaching E. Let c_n be the number of paths of exactly $2n$ jumps ending at C without reaching E. Then c_n is the same number that ends at G by symmetry.

To jump to A in $2n$ steps, the frog may first jump to A in $2n-2$ steps and then jump to A in 2 ways (via B or H), or jump to C or G in $2n-2$ steps and then jump to A in 1 way. This shows

$$b_n = 2b_{n-1} + 2c_{n-1}. \tag{1}$$

Similarly, to jump to C in $2n$ steps, the frog may jump to A in $2n-2$ steps and then jump to C in 1 way, or jump to C in $2n-2$ steps and then jump to C in 2 ways. This shows

$$c_n = b_{n-1} + 2c_{n-1}. \tag{2}$$

Combining (1) and (2), we have

$$c_{n+2} - 2c_{n+1} = b_{n+1} = 2b_n + 2c_n = 2(c_{n+1} - 2c_n) + 2c_n,$$

i.e. $c_{n+2} = 4c_{n+1} - 2c_n$. Its characteristic equation is $\lambda^2 - 4\lambda + 2 = 0$. Its roots are $2 \pm \sqrt{2}$. Thus, we have $c_n = \alpha(2+\sqrt{2})^n + \beta(2-\sqrt{2})^n$ for some constants α, β. One easily finds that $c_0 = 0$ and $c_1 = 1$. Solving the system

$$\begin{cases} \alpha + \beta = 0, \\ \alpha(2+\sqrt{2}) + \beta(2-\sqrt{2}) = 1, \end{cases}$$

we obtain $\alpha = \dfrac{1}{2\sqrt{2}}$ and $\beta = -\dfrac{1}{2\sqrt{2}}$. Hence,

$$c_n = \frac{1}{2\sqrt{2}}((2+\sqrt{2})^n - (2-\sqrt{2})^n).$$

To jump to E in $2n$ steps, the frog may jump to C or G in $2n-2$ steps and then jump to E in 1 way. Thus,

$$a_{2n} = 2c_{n-1} = \frac{(2+\sqrt{2})^{n-1} - (2-\sqrt{2})^{n-1}}{\sqrt{2}} = \frac{1}{\sqrt{2}}(x^{n-1} - y^{n-1}).$$

30. Let b_m be the number of such tuples. When $m = 1$, a_1 can be 0 or n. So $b_1 = 2$.

Consider b_{m+1} where $m \geqslant 1$. If $a_{m+1} = n$, then we need $n \mid a_1 + a_2 + \cdots + a_m$. There are b_m ways to choose the first m entries. If $0 \leqslant a_{m+1} \leqslant n-1$, then there is always a unique way to choose a_{m+1} such that $a_{m+1} \equiv -(a_1 + a_2 + \cdots + a_m) \pmod{n}$ regardless of the values of $a_1, a_2, \ldots, a_m$. Hence,

$$b_{m+1} = b_m + (n+1)^m.$$

Therefore,

$$\begin{aligned} b_m &= b_{m-1} + (n+1)^{m-1} = b_{m-2} + (n+1)^{m-2} + (n+1)^{m-1} \\ &= \cdots = b_1 + (n+1) + (n+1)^2 + \cdots + (n+1)^{m-1} \\ &= 2 + \frac{(n+1)^m - (n+1)}{(n+1) - 1} = \frac{(n+1)^m + n - 1}{n}. \end{aligned}$$

31. Let b_n be the number of such permutations with 2014 replaced by n. Consider any such permutation and let k be the index for which $a_k = 1$. Since $(k-j) + a_{k-j} \leqslant k + a_k = k+1$, we have $a_{k-j} \leqslant j+1$ for each $j = 1, 2, \ldots, k-1$. From $a_{k-1} \leqslant 2$, we have $a_{k-1} = 2$. From $a_{k-2} \leqslant 3$, we have $a_{k-2} = 3$. Inductively, we find that $a_{k-j} = j+1$ for each j.

The remaining terms $a_{k+1}, a_{k+2}, \ldots, a_n$ is a permutation of $k+1, k+2, \ldots, n$. Also, for $i \leqslant k < j$, we must have

$$i + a_i = k + 1 < j + a_j.$$

So only the conditions for which the indices i, j are larger than k are required. If we subtract each of the remaining terms by k, the condition is the same. So there are b_{n-k} ways to permute the remaining numbers, where $b_0 = 1$. Summing over all k's, we get

$$b_n = b_0 + b_1 + \cdots + b_{n-1}.$$

Using $b_0 = 1$ and $b_1 = 1$, we can prove inductively that $b_n = 2^{n-1}$ for $n \geqslant 1$. So the answer is $b_{2014} = 2^{2013}$.

32. Let $|A_n|$ be the number of such permutations. For $n = 1, 2, 3$, we note that the conditions are always satisfied for any permutation. Therefore, we have $|A_1| = 1$, $|A_2| = 2$ and $|A_3| = 6$. Consider $n \geqslant 4$. Since

$$2(a_1 + a_2 + \cdots + a_n) = 2(1 + 2 + \cdots + n) = n(n+1),$$

the divisibility by n is always satisfied. Since we need $2(a_1 + \cdots + a_{n-1}) = n(n+1) - 2a_n$ to be divisible by $n-1$, we have $n-1 \mid 2-2a_n$. For even n, this yields $a_n \equiv 1 \pmod{n-1}$ so that $a_n = 1$ or n. For odd n, this yields $a_n \equiv 1 \pmod{\frac{n-1}{2}}$ so that $a_n = 1, \frac{n+1}{2}$ or n.

We first consider the odd case with $a_n = \frac{n+1}{2}$. From the divisibility by $n-2$, since

$$2(a_1 + \cdots + a_{n-2}) = n(n+1) - (n+1) - 2a_{n-1},$$

we need $n - 2 \mid 3 - 2a_{n-1}$. As n is odd, this gives the unique case $a_{n-1} = \frac{n+1}{2}$ (note that $\frac{n+1}{2} - (n-2) < 1$ and $\frac{n+1}{2} + (n-2) > n$ for $n > 3$). This contradicts the fact that a_{n-1} and a_n should be distinct.

Now, if $a_n = n$, then the number of choices for the remaining numbers is exactly $|A_{n-1}|$ by definition, since $\{a_1, \ldots, a_{n-1}\} = \{1, 2, \ldots, n-1\}$. If $a_n = 1$, then we have $\{a_1, \ldots, a_{n-1}\} = \{2, 3, \ldots, n\}$. In that case, we can reduce each of the numbers $a_1, \ldots, a_{n-1}$ by 1, so that $\{a_1, \ldots, a_{n-1}\} = \{1, 2, \ldots, n-1\}$ and all congruence conditions are the same (as $k \mid 2(a_1 + \cdots + a_k)$ if and only if $k \mid 2((a_1 - 1) + \cdots + (a_k - 1))$). Therefore, the number of choices is again $|A_{n-1}|$.

Therefore, we get a relation $|A_n| = 2|A_{n-1}|$ for $n > 3$. Hence, $|A_n| = 3 \times 2^{n-2}$ for $n \geqslant 4$.

33. We arrange the numbers to form the following table.

1	2	3	$\cdots$	n
$n+1$	$n+2$	$n+3$	$\cdots$	$2n$

Choosing a subset T is the same as choosing some cells in the table such that no two are adjacent and the cells n and $n+1$ are not chosen simultaneously. Let a_n be the number of ways to choose some cells of this table such that no two are adjacent. Let b_n be the number of those ways such that the cell $n+1$ is not chosen. Let c_n be the number of those ways such that the cells n and $n+1$ are both not chosen. We first establish some recurrence relations.

For a_n, if both 1 and $n+1$ are not chosen, then we only need to choose the cells from the remaining $2 \times (n-1)$ table. There are a_{n-1} ways to do so. If 1 is chosen but not

$n+1$, then we cannot choose 2. Removing the cells 1, 2 and $n+1$, we are left with a $2 \times (n-1)$ table with one corner removed. There are b_{n-1} ways to choose the cells in this table. The same number applies if $n+1$ is chosen but not 1. So we have

$$a_n = a_{n-1} + 2b_{n-1}. \tag{1}$$

For b_n, if 1 is not chosen, then we choose the cells from a $2 \times (n-1)$ table. There are a_{n-1} ways to do so. If 1 is chosen, then we cannot choose 2. Again, there are b_{n-1} ways to do so. This yields

$$b_n = a_{n-1} + b_{n-1}. \tag{2}$$

Similarly, for c_n, if 1 is not chosen, then there are b_{n-1} ways to choose the cells. If 1 is chosen, then 2 cannot be chosen. Then either $n+2$ is chosen or not. In the latter case, we are left with the case which yields b_{n-2} ways. In the former case, $n+3$ cannot be chosen, and we are left with the case which yields c_{n-2} ways. Thus, we get

$$c_n = b_{n-1} + b_{n-2} + c_{n-2}. \tag{3}$$

Let s_n be the number of ways to choose the cells which meet our goal. Among the cases counted by a_n, those for which both n and $n+1$ are chosen should be eliminated. In those cases, all of $1, n-1, n+2, 2n$ cannot be chosen. There are c_{n-2} ways to choose the remaining cells. This implies

$$s_n = a_n - c_{n-2}. \tag{4}$$

Firstly, using (1) and (2), we have

$$a_{n-1} = b_n - b_{n-1} = \frac{1}{2}(a_{n+1} - a_n) - \frac{1}{2}(a_n - a_{n-1}),$$

which means

$$a_{n+1} = 2a_n + a_{n-1}.$$

Since the roots of the characteristic polynomial $\lambda^2 - 2\lambda - 1 = 0$ are $1 \pm \sqrt{2}$, and the initial conditions are $a_1 = 3$, $a_2 = 7$, we easily obtain

$$a_n = \frac{(1+\sqrt{2})^{n+1} + (1-\sqrt{2})^{n+1}}{2}.$$

Then from (1) we get

$$\begin{aligned} b_n &= \frac{1}{2}(a_{n+1} - a_n) = \frac{(1+\sqrt{2})^{n+2} + (1-\sqrt{2})^{n+2} - (1+\sqrt{2})^{n+1} - (1-\sqrt{2})^{n+1}}{4} \\ &= \frac{(1+\sqrt{2})^{n+1} - (1-\sqrt{2})^{n+1}}{2\sqrt{2}}. \end{aligned}$$

Now, (3) becomes

$$\begin{aligned} c_n &= c_{n-2} + \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}} + \frac{(1+\sqrt{2})^{n-1} - (1-\sqrt{2})^{n-1}}{2\sqrt{2}} \\ &= c_{n-2} + \frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2}. \end{aligned}$$

It is easy to deduce

$$\begin{aligned} c_n &= c_{n-2} + \frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2} \\ &= c_{n-4} + \frac{(1+\sqrt{2})^{n-2} + (1-\sqrt{2})^{n-2}}{2} + \frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2} \\ &= \cdots \\ &= c_r + \frac{(1+\sqrt{2})^{r+2} + (1-\sqrt{2})^{r+2}}{2} + \cdots + \frac{(1+\sqrt{2})^{n-2} + (1-\sqrt{2})^{n-2}}{2} \\ &\quad + \frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2} \\ &= c_r + \frac{(1+\sqrt{2})^{n+1} - (1+\sqrt{2})^{r+1} + (1-\sqrt{2})^{n+1} - (1-\sqrt{2})^{r+1}}{4} \end{aligned}$$

where $r = 1$ if n is odd and $r = 2$ if n is even. Since $c_1 = 1$ and $c_2 = 4$, we find that

$$c_n = \frac{(1+\sqrt{2})^{n+1} + (1-\sqrt{2})^{n+1} + 2\cdot(-1)^n}{4}.$$

Finally, from (4) we have

$$\begin{aligned} s_n &= \frac{(1+\sqrt{2})^{n+1} + (1-\sqrt{2})^{n+1}}{2} - \frac{(1+\sqrt{2})^{n-1} + (1-\sqrt{2})^{n-1} + 2\cdot(-1)^{n-2}}{4} \\ &= \frac{(3+\sqrt{2})(1+\sqrt{2})^n + (3-\sqrt{2})(1-\sqrt{2})^n - 2\cdot(-1)^n}{4}. \end{aligned}$$

Bijection

34. We rearrange 30 objects together with 3 sticks in a row. The sticks partition the objects into 4 regions. The number of objects in the 4 regions correspond to the variables a, b, c, d respectively. Such a correspondence gives a bijection between the nonnegative integer solutions to the equation $a + b + c + d = 30$ and the rearrangements of the objects and sticks. Since there are $\binom{30+3}{3} = 5456$ ways to permute the things, the number of solutions is 5456.

35. We put 30 objects in a row and we put 3 sticks in different space between two objects. The sticks partition the objects into 4 nonempty regions. The number of objects in the 4 regions correspond to the variables a, b, c, d respectively. Such a correspondence gives a bijection between the positive integer solutions to the equation $a+b+c+d=30$ and the placement of the sticks. Since there are $\binom{29}{3} = 3654$ ways to place the sticks, the number of solutions is 3654.

36. Let
$$a' = a-3,\ b' = b-3,\ c' = c-3,\ d' = d-3.$$
The equation becomes $a'+b'+c'+d' = 18$ and the condition becomes $0 \leqslant a', b', c', d' \leqslant 6$. Firstly, using the bijection method, we find that the number of nonnegative integer solutions to $a'+b'+c'+d' = 18$ is $\binom{18+4-1}{4-1} = 1330$. We need to eliminate those solutions for which some variables are larger than 6.

Consider the case when $a' \geqslant 7$ and $b', c', d' \geqslant 0$. Let $a'' = a'-7$. The equation becomes $a''+b'+c'+d' = 11$. The number of nonnegative integer solutions is $\binom{11+4-1}{4-1} = 364$. By symmetry, the number of nonnegative integer solutions to $a'+b'+c'+d' = 18$ with $b' \geqslant 7$ (or $c' \geqslant 7$ or $d' \geqslant 7$) is also 364.

Next, consider the case when two of the variables, say a' and b' are at least 7. Let $a'' = a'-7$ and $b'' = b'-7$. The equation becomes $a''+b''+c'+d' = 4$. The number of nonnegative integer solutions is $\binom{4+4-1}{4-1} = 35$.

Note that it is not possible for at least three of a', b', c', d' are at least 7. Therefore, by the inclusion-exclusion principle, the number of solutions satisfying the condition is
$$1330 - 4 \times 364 + \binom{4}{2} \times 35 = 84.$$

37. Let $e = 30-a-b-c-d$. Then e is a nonnegative integer and we have $a+b+c+d+e = 30$. Conversely, any solution to $a+b+c+d+e = 30$ gives a unique and distinct tuple (a, b, c, d) of nonnegative integers such that $a+b+c+d \leqslant 30$. This gives a bijection between the solutions to the inequality and the solutions to the Diophantine equation. From the standard method, we find that there are $\binom{30+5-1}{5-1} = 46376$ desired tuples.

38. Consider $f(m, n)$. Suppose there are k of the x_j's equal to 0. There are $\binom{n}{k}$ ways to choose the 0's. WLOG assume $x_1, x_2, \ldots, x_{n-k}$ are nonzero. Then it suffices to count the number of solutions to the inequality
$$x_1 + x_2 + \cdots + x_{n-k} \leqslant m$$

in positive integers and then multiplied the result by 2^{n-k} by assigning $\pm$ signs to the variables. Each solution corresponds to a positive integer solution to

$$x_1 + x_2 + \cdots + x_{n-k} + y = m + 1.$$

There are $\binom{m}{n-k}$ solutions by the standard method. Therefore,

$$f(m,n) = \sum_{k=0}^{n} 2^{n-k}\binom{n}{k}\binom{m}{n-k} = \sum_{\ell=0}^{n} 2^{\ell}\binom{n}{\ell}\binom{m}{\ell}$$

by using the change of variable $\ell = n-k$ and the fact $\binom{n}{k} = \binom{n}{n-k}$.

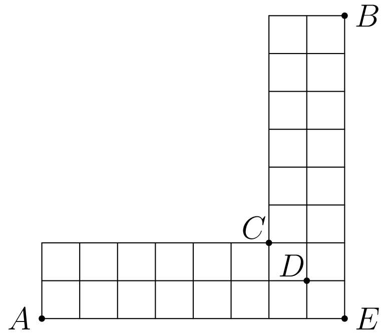
WLOG assume $m \geqslant n$. Then

$$f(n,m) = \sum_{\ell=0}^{m} 2^{\ell}\binom{m}{\ell}\binom{n}{\ell} = \sum_{\ell=0}^{n} 2^{\ell}\binom{m}{\ell}\binom{n}{\ell} = f(m,n)$$

since $\binom{n}{\ell} = 0$ if $\ell > n$.

39. No matter how we move, we need to move 8 steps upwards and 8 steps rightwards in total. Denote by U and R an upward step and a rightward step respectively. Thus, any path corresponds to a rearrangement of 8 copies of U's and 8 copies of R's. Conversely, any such rearrangement corresponds to a unique and distinct path. Thus, the number of paths is $\binom{8+8}{8} = 12870$.

40. Any path from A to B must pass through exactly one of C, D, E as shown. Firstly, a path from A to B through C can be decomposed into a path from A to C and a path from C to B. Each of these corresponds to a choice of path from a 2×6 board. Using the bijection method, there are $\binom{2+6}{2} = 28$ choices for each path. Thus, there are $28^2 = 784$ paths passing through C.

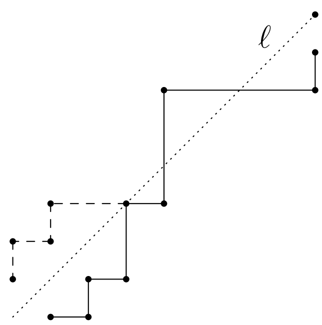


Similarly, there are $\binom{1+7}{1}^2 = 64$ paths passing through D, and obviously 1 path passing through E. In total, there are $784 + 64 + 1 = 849$ paths from A to B.

41. We need to walk $2n$ steps in total. There are $\binom{2n}{n}$ paths if the last condition is ignored since we need to choose n rightwards steps from $2n$ steps. It remains to find the number of paths which meet the line $\ell : y = x + 1$.

    For each path that meets the line $y = x + 1$, we reflect the part of the path from $(0, 0)$ to the first intersection point of the path and ℓ in the line ℓ. This gives a new path from $(-1, 1)$ to (n, n). Conversely, for any path from $(-1, 1)$ to (n, n), there must be an intersection point with ℓ. So we can always do the same reflection to obtain a path from $(0, 0)$ to (n, n) that meets ℓ. Such a correspondence is clearly a bijection. So it is the same as counting the number of paths from $(-1, 1)$ to (n, n). There are $\binom{2n}{n+1}$ paths. Hence, the answer is

$$\binom{2n}{n} - \binom{2n}{n+1} = \binom{2n}{n} - \frac{n}{n+1}\binom{2n}{n} = \frac{1}{n+1}\binom{2n}{n}.$$



42. Consider those n-digit numbers except $44\cdots4$. Each of them contains a digit which is distinct from 4. We choose the last such digit. If this digit is 1, then the n-digit number corresponds to two other numbers by replacing this 1 with 2 and 3 respectively. In this way, we can partition the $4^n - 1$ numbers into groups of 3, such that the numbers in the same group only differ by a single digit. Note that the 3 numbers in the same group leave different remainders when divided by 3 since the remainders can be found by adding all the digits, and $1, 2, 3$ leave different remainders when divided by 3.

    Therefore, if n is a multiple of 3, then $44\cdots4$ is a multiple of 3, and hence there are $\frac{4^n - 1}{3} + 1 = \frac{4^n + 2}{3}$ such numbers. If n is not a multiple of 3, then there are $\frac{4^n - 1}{3}$ such numbers.

43. To form a subset of $\{1,2,\ldots,n\}$, each of the n elements can be chosen or not. If an element m is chosen, then it contributes $\frac{1}{m}$ to the product. If it is not chosen, then it contributes 1 to the product. Therefore, each summand corresponds to a term in the expansion of

$$\left(1+\frac{1}{1}\right)\left(1+\frac{1}{2}\right)\left(1+\frac{1}{3}\right)\cdots\left(1+\frac{1}{n}\right).$$

For example, if we pick the subset $\{1,2,4,5\}$, then the corresponding summand is $\frac{1}{1\cdot 2\cdot 4\cdot 5}$. This can be obtained by choosing $\frac{1}{1},\frac{1}{2},\frac{1}{4},\frac{1}{5}$ in the brackets and 1's from the other brackets. However, the product 1 obtained by choosing 1's from all brackets corresponds to the empty set, which should be removed. Thus, the sum equals to

$$\left(1+\frac{1}{1}\right)\left(1+\frac{1}{2}\right)\cdots\left(1+\frac{1}{n}\right)-1=\frac{2}{1}\times\frac{3}{2}\times\cdots\times\frac{n+1}{n}-1=(n+1)-1=n.$$

44. We prove that there is a bijection between the partitions of the set $S=\{1,2,\ldots,n\}$ satisfying the conditions and the partitions of S into blocks of consecutive integers (e.g. $(1,2),(3),(4,5,6,7),(8),(9,10)$ is a partition of $\{1,2,\ldots,10\}$). Given each partition of S into blocks of consecutive integers, say

$$(1,2,\ldots,m_1),(m_1+1,\ldots,m_2),(m_2+1,\ldots,m_3),\ldots,(m_{t-1}+1,\ldots,n),$$

we form A_1,A_2,A_3 as follows. Firstly, we put $1,2,\ldots,m_1$ into a set, say A_1. Then we put $m_1+1,\ldots,m_2$ into another set, say A_2. Suppose we have put the integers from 1 to m_j into some sets, where m_j is put in A_k for some $k=1,2,3$. After that, we put $m_j+1,\ldots,m_{j+1}$ into the same subset different from A_k. We claim that there is a unique choice of this subset for which the conditions are satisfied. It suffices to prove that the largest numbers in the two subsets other than A_k have different parities. (If A_3 is empty, then the parity of the smallest number which may be put in A_3 is uniquely determined from the condition, and hence we can still assign a parity to A_3.)

This can be proved inductively. The base case is the situation when we put the block $(m_2+1,\ldots,m_3)$. If m_1 is odd, then the smallest number which may be put in A_3 is odd, and so m_2+1 can only be put in exactly one of A_1 and A_3. Similarly, if m_1 is even, then the smallest number which may be put in A_3 is even and the same holds. Suppose we now put the block $(m_j+1,\ldots,m_{j+1})$. Since $(m_{j-1}+1,\ldots,m_j)$ is the previous block put in A_k, the largest number in another set is m_{j-1}, and the largest number in the remaining set has the same parity as $m_{j-1}+1$ by the inductive hypothesis, and hence different parities as m_{j-1}. This is what we need.

Therefore, there is always a unique way to put the blocks into the subsets so that the conditions are satisfied. Conversely, every partition of S into three subsets not in order satisfying the conditions corresponds to a unique way of partition of S into blocks in the obvious way (i.e. the first block is $(1,2,\ldots,m_1)$ where m_1 is the largest number such that $1, 2, \ldots, m_1$ belong to the same subset, the second block is $(m_1 + 1, \ldots, m_2)$ where m_2 is the largest number such that $m_1 + 1, \ldots, m_2$ belong to the same subset, etc.).

It remains to count the number of partitions of S into blocks of consecutive integers. We simply need to put some sticks in the space of some pairs of consecutive integers to indicate the separations of the blocks. There are $n - 1$ places for which we can put a stick or not. So there are 2^{n-1} partitions. As the subsets are indeed different, there are $3! \times 2^{n-1} = 3 \times 2^n$ partitions in total.